

UNIVERSITY PARTNER



4CS016 – Embedded Systems Programming

Smart Street Lighting System Using Arduino UNO and IR Sensors

Student Name : Karan Sah

University ID : 2329860

Module Leader : Jeffrey Ting

Lecturer Name : Raj Pradhan

Date of Submission : 2023/05/21

Acknowledgement

I would like to thank our module leader Jeffrey Ting for providing with wonderful slides which were immensely helpful in learning embedded systems programming. I would also like to express my thanks to our lecturer Raj Pradhan sir for clarifying further about the embedded systems programming and about the projects. I would also like to thank Sulove Bhattarai sir for helping immensely during tutorials and workshops and aiding through all the simple and complex questions.

My appreciation also is towards my college, my teachers and my classmates who taught and guided me during my studies and project planning. My deepest gratitude for offering suggestions and ideas towards the projects.

Table of Contents

1. Introduction.....	1
2. Tools Used:.....	2
2.1 Fritzing:	2
2.2 Arduino Uno	2
2.3 Infrared Sensors	3
2.4 LEDs	3
2.5 DC Jack.....	3
2.6 Battery	4
2.7 Battery Cap	4
2.8 Jumping Wires.....	5
2.9 Arduino Software and Arduino IDE	5
2.10 Miscellaneous.....	6
3. Working Principle and Procedure.....	6
3.1 Infrared Sensor Detection and Object Detection	6
3.2 Signal Processing	7
3.3 Decision Making	7
3.4 Activating Light	7
3.5 Control of Duration.....	7
4. Circuit Diagram and Fritzing Schematic	8
5. Program Code.....	10
.....	10
6. Testing.....	11
6.1 Phase 1: Basic Structure	12
6.2 Phase 2: All Connected	13
6.3 Phase 3: Test Run	13
6.4 Phase 4: Reflection.....	14
7. Conclusion.....	16

Table of Figures

Figure 1Fritzing	2
Figure 2Arduino Uno	2
Figure 3 Infrared Sensor	3
Figure 4 White LED	3
Figure 5 DC Jack	4
Figure 6 9V Battery	4
Figure 7Battery Cap	5
Figure 8Jumping Wires	5
Figure 9Arduino Software	6
Figure 10 Working of IR Sensor (Robu.in , 2020)	7
Figure 11 Circuit Diagram Showing the Connection between Arduino, Sensors, LEDs and Resistors	8
Figure 12 Fritzing Schematic Showing the Connection between Arduino, Sensors, LEDs and Resistors	9
Figure 13 Program Code--1	10
Figure 14 Program Code Continued- 2	11
Figure 15 Program Code Continued-3	11
Figure 16 Code Compilation	12
Figure 17 Code Upload	12
Figure 18 Basic Structure Completed	12
Figure 19 Connections Completed	13
Figure 20 Error During Testing Phase	14
Figure 21 Successful on the First Sensor and LED	14
Figure 22 Successful on the Second Sensor and LED	15
Figure 23 Successful on the Third and Last Sensor and LED	15

1. Introduction

Street lights have been an essential component of night life in urban and now in rural areas as well. The lights can be seen in highways, asphalt road, rural roads, narrow ways, and footpaths. Basically, there are street lights where there is flow of people and on the side of almost every road.

The benefits of illuminating street lights are numerous. The lighting of street lights aid in reducing night time accidents and crashes through improved visibility. The lights help in navigation and avoid pedestrian crashes by nearly 50%. These lights also make people feel safe as well as the crime rates too can be reduced. Likewise, route lighting aid in decreasing the glares from headlight of vehicles (Road Safety Toolkit, 2022).

Even though these lights are essential, it is found that the traditional street lights (i.e. switching it around evening through switch and leaving it till morning) are not good due to having complications. Some of them being high cost of maintaining the lights throughout the night. The estimated UK's annual bill being around £220M. Also, artificial light at night has huge impact on wildlife and natural world. These can stop the growth of frogs and toads which in turn decrease their egg laying chances (Uttley, et al., 2018). Also, lighting street lights prevent the pollinating processes done by moths (Macgregor, 2016).

Due to these reasons, street lights being a necessity but also it should not be activated all night even though there is no traffic or people at night. Thus came the idea of smart street with automatic motion sensing and lighting. The light automation has huge merits such as saving energy and saving electricity by automatically turning the lights off when not in use, increases price transparency, energy saved during production and making the energy industry more efficient (Matthews, 2019).

Therefore, keeping these ideas in consideration, the following project is similar to the idea and can be considered mini project of smart street light whereby the sensor detects infrared radiation which can be further implemented into sensing proximity, detecting motion and sensing ambient light. In this project, the street lights remain off until and unless there is motion detected in road be it may from vehicles or people and after the motion stops i.e. after vehicles pass through or people finish waking the section of the road the street light automatically turns off.

2. Tools Used:

2.1 Fritzing:

An open source software which is generally used to cross the road of physical and digital components and factors of electronics. This software is basically used to share and create electronic circuit designs, breadboards designs, PCB, write codes on electronics. It is used in creation of virtual circuits and prototypes of different circuits and projects. Fritzing has been used in this project to draw both graphical representation of the circuit through breadboard designing and schematic diagram through schematic designing aspects of the software.



Figure 1Fritzing

2.2 Arduino Uno

A microcontroller board based on the ATmega328P, Arduino Uno comprises of 14 digital output and input pins, analog inputs of 6, ceramic resonator of 16MHz etc. It is an open source programmable board of microcontroller which is used in various range of activities (Arduino Store, 2021). In this project the microcontroller board, Arduino Uno can be considered as one of the main factors.

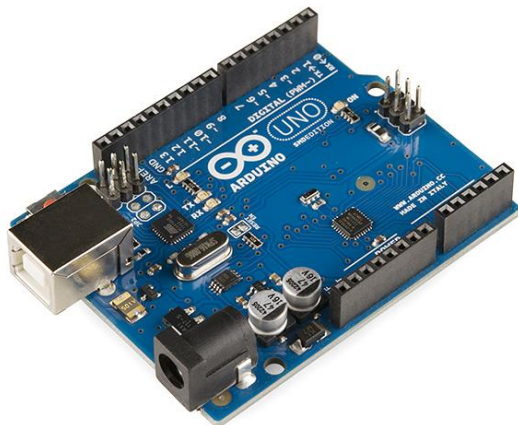


Figure 2Arduino Uno

2.3 Infrared Sensors

The device which measures as well as detects infrared radiation in and around the environment is Infrared Sensor (Jost, 2019). This sensor can be considered as proximity sensor. The IR LED releases a pulse of IR Light and after being reflected by object the pulse is then detect by photodiode and thus generating a signal. Its components are: resistor, photodiode, infrared LED, IC, LED, etc.

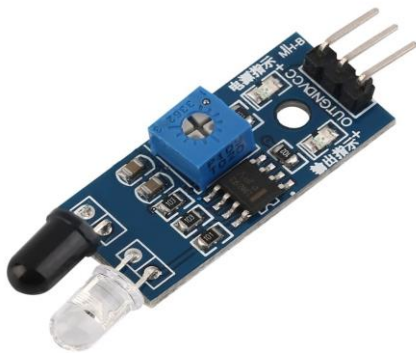


Figure 3 Infrared Sensor

2.4 LEDs

Light-Emitting Diodes, a semiconductor device which emit light after an electric current is passed through are LEDs. These light are uses for various purposes, some being in indicators, displays, communication of optoelectronic (Eda Gungor, 2020).



Figure 4 White LED

2.5 DC Jack

It is a barrel connector or a power jack used to provide direct current to different electronic devices. It has two sides both positive and negative, center pin being the

positive one whereas the outer barrel is the negative one. They are used in laptops, video and audio equipment, routers etc. (H. Z. Khatatneh, 2016).



Figure 5 DC Jack

2.6 Battery

The battery here is used for power source to supply the power. A 9V batter is used here.



Figure 6 9V Battery

2.7 Battery Cap

This cap is used here for connecting battery with the circuit.

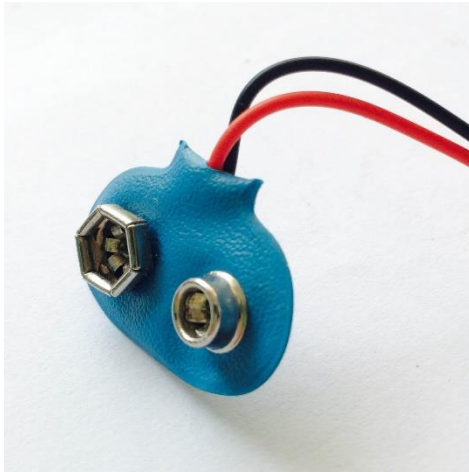


Figure 7 Battery Cap

2.8 Jumping Wires

Different types of Jumping Wires, such as both sides being males, one side male and another side female, and both side female jumping wires are used in this project. These wires have different colors although the colors doesn't possess much meaning but it becomes quite helpful in differentiating different wires used in different places due to its varying colors.

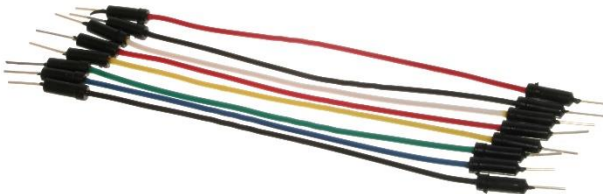


Figure 8 Jumping Wires

2.9 Arduino Software and Arduino IDE

This is an open source tool to program and upload code in Arduino microcontrollers. The interface is simple and the software itself comes with many helpful libraries.



Figure 9 Arduino Software

2.10 Miscellaneous

Other objects that were helpful for the projects were straws, glues, cardboards, bottle caps for adjusting the wires and setting up the environment.

3. Working Principle and Procedure

The working principle behind the project can be categorized into several steps:

3.1 Infrared Sensor Detection and Object Detection

In IR sensor, there comprises of two major components which are mainly responsible for detection, first one being IR LED, the white one and the second one being Photodiode, the black one. The IR LED first transmits IR light to an object which in turn reflects the light to photodiode. The output voltage and resistance of photodiode alters with the proportion to the IR light received by the diode (Robu.in , 2020).

The detection of the infrared radiation takes place when a vehicle or a person enters the proximity of the sensor's detection range. A signal is triggered which indicates that either pedestrian or a vehicle is present in the vicinity. This is done through the IR sensors placed on the side of road strategically.

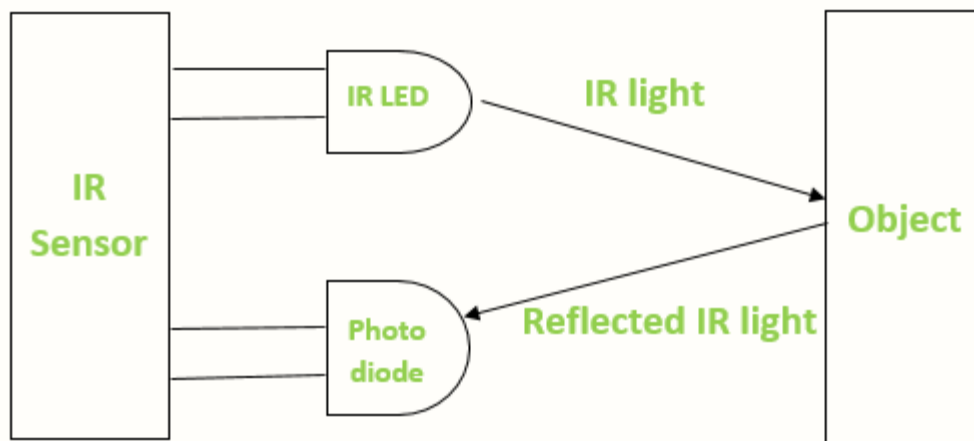


Figure 10 Working of IR Sensor (Robu.in , 2020)

3.2 Signal Processing

The microcontroller of CPU (Central Processing Unit) is responsible to process the signal received from IR sensor in the smart street light system. The analyzation mainly is emphasized whether the object is being detected or not and the location of the object in respect to street light.

3.3 Decision Making

The decision to whether turn on the street light or not is basically made by analyzing signal. After the analyzation of signal is completed, the decision to activate the corresponding street light rests on microcontroller. The light is turned on only when the presence of an object is detected. This works along the range of the sensor.

3.4 Activating Light

Microcontroller sends signal to lighting control circuitry after the decision to activate the street light is made. The circuitry which is responsible for controlling the power supply to the street light, illuminates the street light after receiving the signal around the area of object detection.

3.5 Control of Duration

After any pedestrian or vehicle has left the proximity of the sensor then the light automatically turns off. Thus, giving the pedestrian or vehicle enough time to pass

through in illumination and not leaving them hanging in the dark. Thereafter, the motion is detected by nearby sensor which in turn turns on the light thus continuing the process of lighted road along the ride or walk during night saving both energy and cost.

Steps:

- LED1, LED2, LED3's positive terminal are connected to 220 ohm resistor and the other end of resistor is connected to pin 5, 6, 7 as output. Whereas, the negative terminals of all three LEDs are grounded.
- The VCCs of all three IR Sensors are connected to 5V in Arduino while all three GNDs of IR sensors are grounded. Likewise, OUT of all three IR Sensors are connected to pin 2, 3, and 4 as input.
- The program is written such that the different scenarios have different results.

4. Circuit Diagram and Fritzing Schematic

Karan Sah
2329860

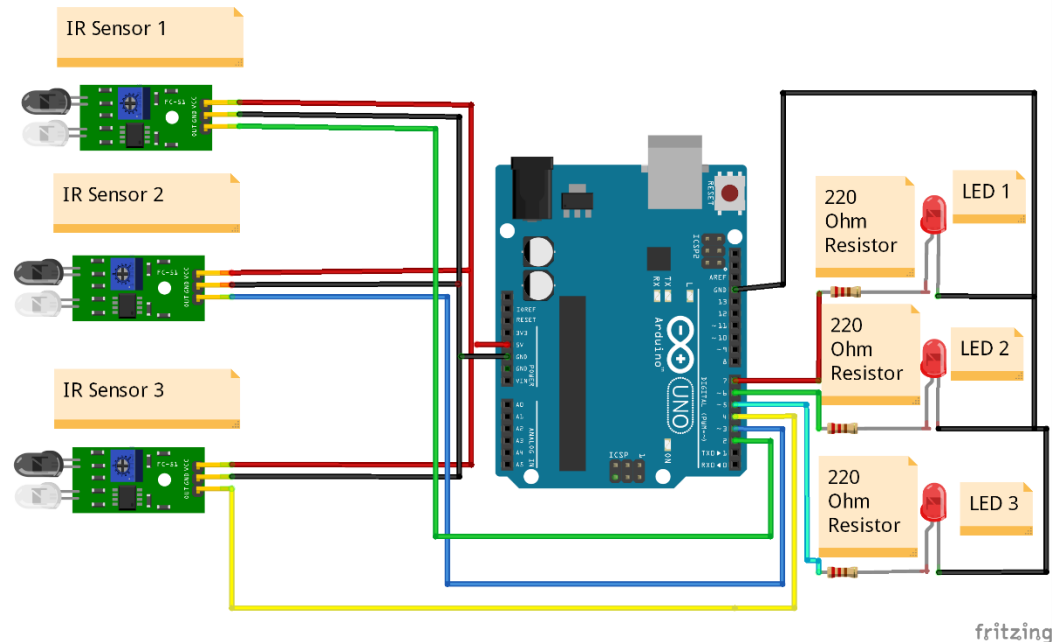
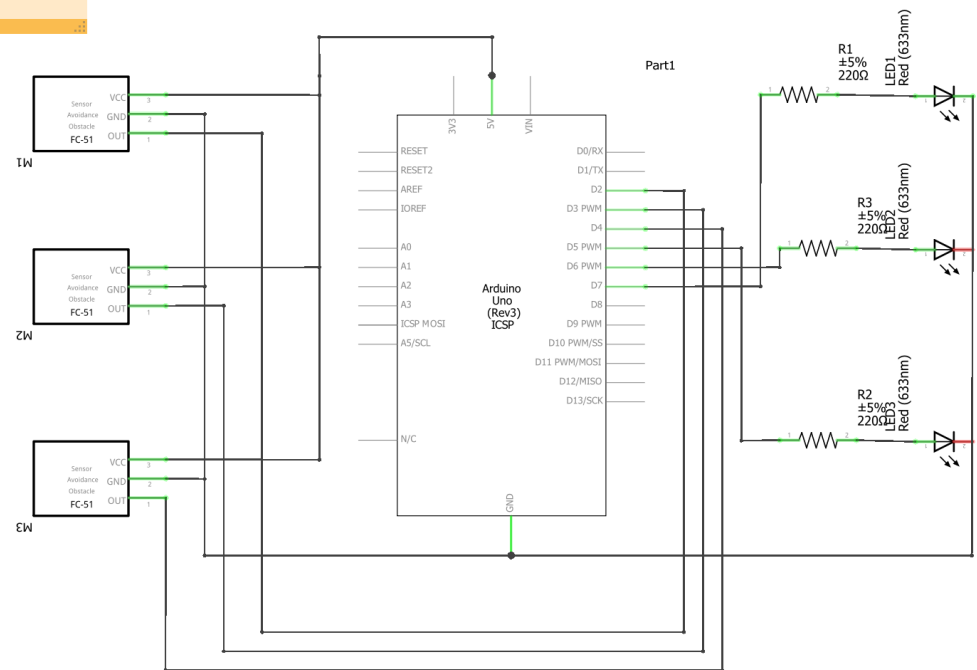


Figure 11 Circuit Diagram Showing the Connection between Arduino, Sensors, LEDs and Resistors



fritzing

Figure 12 Fritzing Schematic Showing the Connection between Arduino, Sensors, LEDs and Resistors

5. Program Code

```
//Karan Sah
//2329860
//Date: 2023/05/21
//Smart Street Lighting System Using Arduino UNO and IR Sensors
// Arduino Program

void setup() {
  pinMode(2, INPUT); // Configure pin 2 as input for IR Sensor 1
  pinMode(3, INPUT); // Configure pin 3 as input for IR Sensor 2
  pinMode(4, INPUT); // Configure pin 4 as input for IR Sensor 3
  pinMode(5, OUTPUT); // Configure pin 5 as output for LED 1
  pinMode(6, OUTPUT); // Configure pin 6 as output for LED 2
  pinMode(7, OUTPUT); // Configure pin 7 as output for LED 3
}

void loop() {
  // Check the state of IR sensors and control LEDs accordingly

  if (digitalRead(2) == LOW && digitalRead(3) == LOW && digitalRead(4) == LOW) {
    // If all IR sensors are LOW (object detected)
    digitalWrite(5, HIGH); // Turn on LED 1
    digitalWrite(6, HIGH); // Turn on LED 2
    digitalWrite(7, HIGH); // Turn on LED 3
  }
  else if (digitalRead(2) == LOW && digitalRead(3) == LOW) {
    // If IR Sensor 1 and IR Sensor 2 are LOW
    digitalWrite(5, HIGH); // Turn on LED 1
    digitalWrite(6, HIGH); // Turn on LED 2
    digitalWrite(7, LOW); // Turn off LED 3
  }
}
```

Figure 13 Program Code--1

```

else if (digitalRead(3) == LOW && digitalRead(4) == LOW) {
    // If IR Sensor 2 and IR Sensor 3 are LOW
    digitalWrite(5, LOW); // Turn off LED 1
    digitalWrite(6, HIGH); // Turn on LED 2
    digitalWrite(7, HIGH); // Turn on LED 3
}
else if (digitalRead(2) == LOW && digitalRead(4) == LOW) {
    // If IR Sensor 1 and IR Sensor 3 are LOW
    digitalWrite(5, HIGH); // Turn on LED 1
    digitalWrite(6, LOW); // Turn off LED 2
    digitalWrite(7, HIGH); // Turn on LED 3
}
else if (digitalRead(2) == LOW) {
    // If only IR Sensor 1 is LOW
    digitalWrite(5, HIGH); // Turn on LED 1
    digitalWrite(6, LOW); // Turn off LED 2
    digitalWrite(7, LOW); // Turn off LED 3
}
else if (digitalRead(3) == LOW) {
    // If only IR Sensor 2 is LOW
    digitalWrite(5, LOW); // Turn off LED 1
    digitalWrite(6, HIGH); // Turn on LED 2
    digitalWrite(7, LOW); // Turn off LED 3
}
else if (digitalRead(4) == LOW) {
    // If only IR Sensor 3 is LOW
    digitalWrite(5, LOW); // Turn off LED 1
    digitalWrite(6, LOW); // Turn off LED 2
    digitalWrite(7, HIGH); // Turn on LED 3
}
}

```

Figure 14 Program Code Continued- 2

```

else {
    // If no IR sensors are LOW (no object detected)
    digitalWrite(6, LOW); // Turn off LED 2
    digitalWrite(7, LOW); // Turn off LED 3
    digitalWrite(5, LOW); // Turn off LED 1
}
}

```

Figure 15 Program Code Continued-3

6. Testing

When code is compiled, the IDE is done compiling and shows no mistake:

Output

```
Sketch uses 1152 bytes (3%) of program storage space. Maximum is 32256 bytes.  
Global variables use 9 bytes (0%) of dynamic memory, leaving 2039 bytes for local variables. Maximum is 2048 bytes.
```

Figure 16 Code Compilation

When uploading, the code is uploaded and shows no errors.

Output

```
Sketch uses 1152 bytes (3%) of program storage space. Maximum is 32256 bytes.  
Global variables use 9 bytes (0%) of dynamic memory, leaving 2039 bytes for local variables. Maximum is 2048 bytes.
```

Figure 17 Code Upload

6.1 Phase 1: Basic Structure

Here the basic structure of the project is completed, where the green cardboard can be considered as a garden and black cardboard considered as road. There are three street light on one side of road. Simple wirings also has been completed.

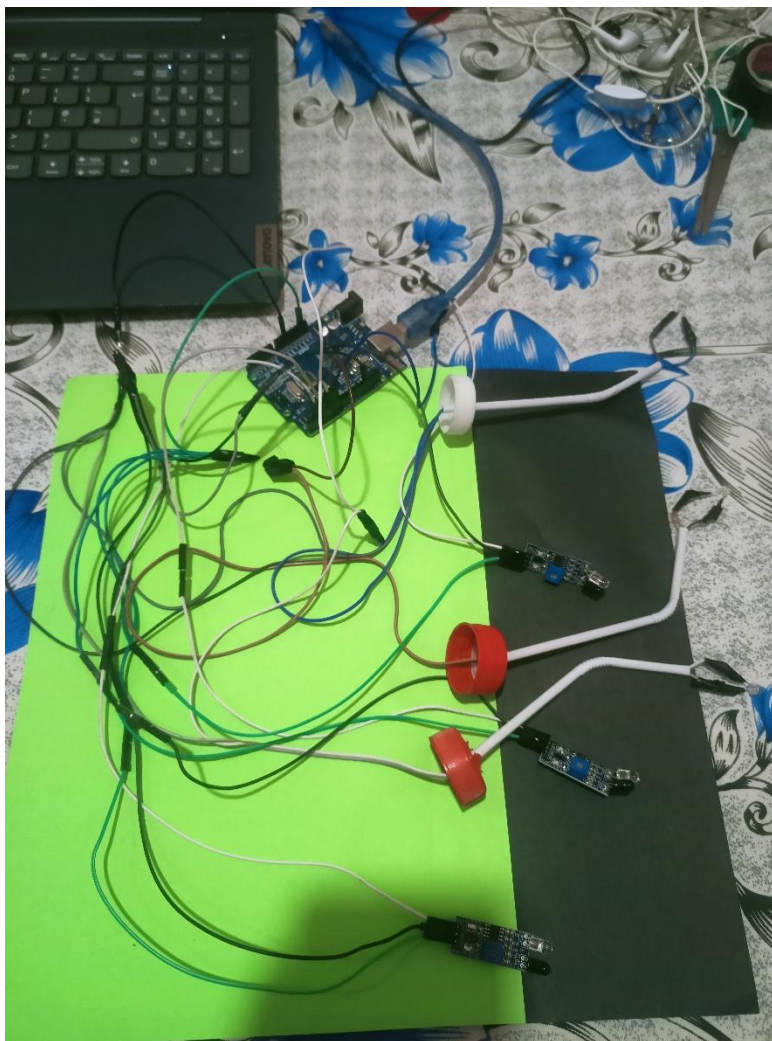


Figure 18 Basic Structure Completed

6.2 Phase 2: All Connected

All the wires have been connected to their respective pins as written in code. The negative terminals of very LEDs are joined together in a resistor and the resistor is grounded in Arduino. The positive terminal of three LEDs are separately bounded to male jumping wires. Likewise, the VCCs of three IR Sensors are bounded together and put in 5V of Arduino. The GND of the sensors are bundled together and grounded in Arduino. The out of each sensors are bounded separately to a male jumping wire.

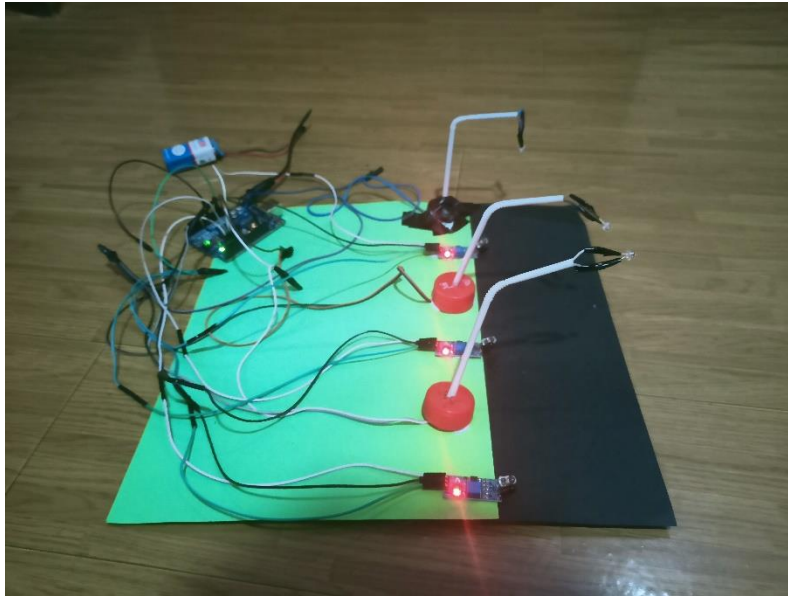


Figure 19 Connections Completed

6.3 Phase 3: Test Run

There were definitely some errors during the testing phase but it was quickly solved by rotating the top which in turn send IR light farther thus solving the problem of the LED not turning on when the vehicles are farther than the sensors. Likewise both the IR LED and photodiode were not working but on further inspection, I found out that both the IR LED and photodiode has to be slightly above the cardboard, not touching them and not being too above the cardboard as well.

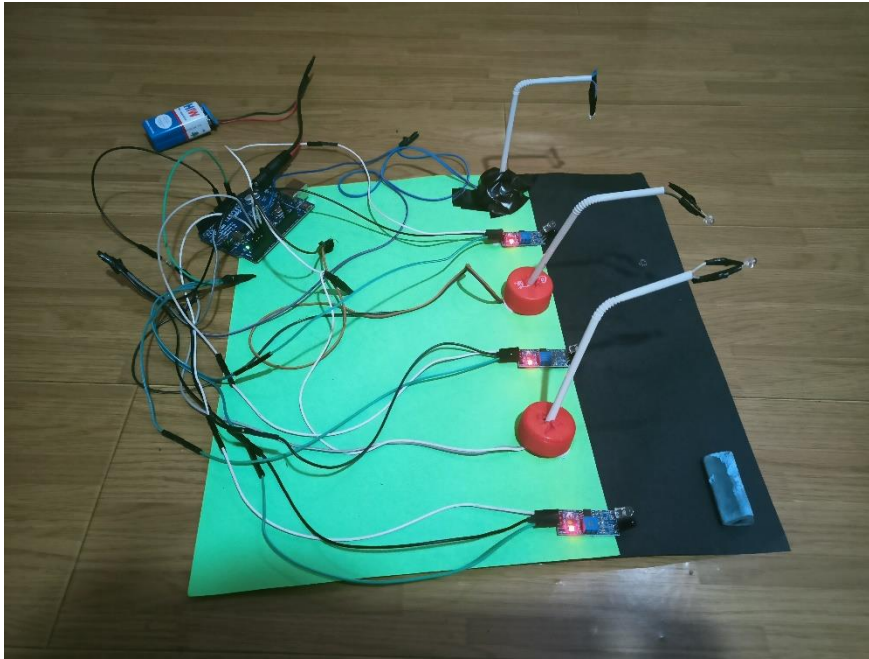


Figure 20 Error During Testing Phase

6.4 Phase 4: Reflection

As a pedestrian or a vehicle moves along the road the IR light is reflected to photodiode which in turn passes the signal to microcontroller and afterwards going through the lighting circuitry, the LED turns on. And when a pedestrian or a vehicle passes away from the sensor, and as the sensor is a proximity sensor, after going away from the sensor's range the LED turns off in turn saving a lot of energy and electric costs, which in turn also increases the life span of LEDs too. The light works on both vehicle and pedestrian thus, making the nights for pedestrian and vehicle drivers much safer from car accidents and also from less crime due to bright night along with saving electricity and energy.

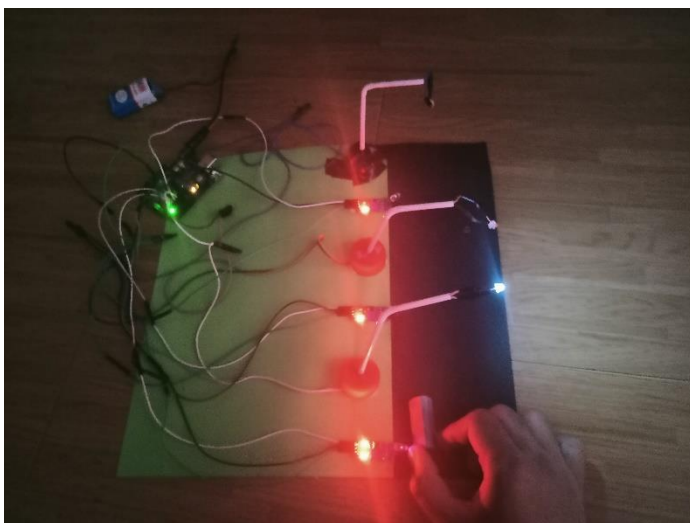


Figure 21 Successful on the First Sensor and LED

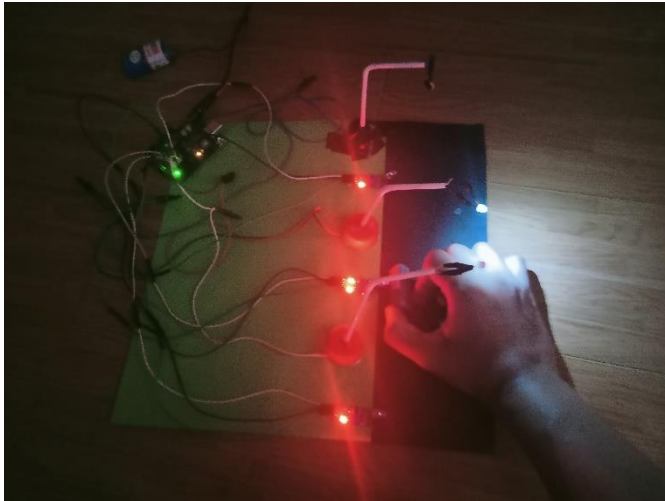


Figure 22 Successful on the Second Sensor and LED

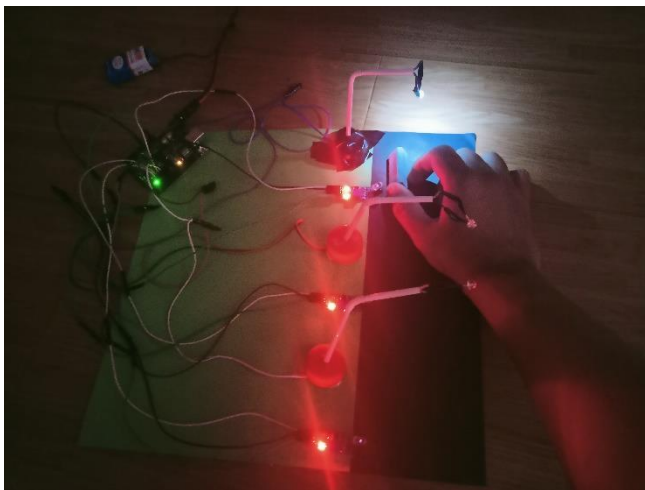


Figure 23 Successful on the Third and Last Sensor and LED

After the errors above were handled, the project ran smoothly. The project was very fun to me because it was definitely very practical in today's world. The thoughts of energy crisis and energy conservation are increasing. And having these kinds of automatic lights off and on systems not only in street lights but in other field will most certainly contribute to the cause of energy conservation. At first, during just the start of the project, the whole project was in mess. As there were a lot of components and definitely a lot of wire both the wires and components would tangle together. Nevertheless, the project did give me a chance to learn new thing, accomplish a very interesting project all by myself. This project certainly was a huge step toward creating and researching. Moreover, this opportunity of doing this project most definitely has been learning experience such as the parts shop, the creativity parts of finding straws and bottle caps, making it stand without glues etc. This project has inspired me to create more and do more of such or entirely different project in the coming future.

7. Conclusion

The report depicts details on street light and why IR sensor light that can also be called automatic lights need to be used in compared to its traditional street light. The report states the advantages of automatic light in respect to its traditional parts and the advantages are huge. The smart street lighting system is made though different components and coding in Arduino IDE. The components include fritzing, Arduino UNO, IR Sensors, 3 White LEDs, DC cable, a 9V Battery, Resistors, Battery Cap, Jumping Wire, Arduino Software and IDE and other miscellaneous objects. The project works in the way such that when IR light sent by IR LED is reflected back by an object it is then sent to photodiode which in turn sends the signal to microprocessor which makes the decision to turn on the light. Thereafter the signal is sent to lighting circuitry which in turn lights the LED. The working mechanism basically takes places in five steps, them being Infrared Sensor Detection and Object Detection, Signal Processing, Decision Making, Activating Light and Control of Duration that being until the proximity of the sensor or the duration could also be extended through different means. The circuit diagram and fritzing schematic is also clearly shown in the project which can also be considered as prototype of the project which was very helpful during actual project making. The code is written in Arduino IDE then uploaded to microcontroller of Arduino. The test is done so that when the car is moving the sensor detects the IR being reflected back and lights up the LED. Moreover, after the objects moves away from the proximity of the sensor, the LED is once again switched off. Likewise, when the object moves forward the IR reflected back by the object is detected by the another IR sensor and the process repeats itself again and again.

This project can be considered as one of the essential project to be implemented in not only urban but also rural areas. As the project is both cost effective and has massive advantage when comes to energy conservation, the project can be considered as boon not only in country Nepal but all over the world.

References

- Arduino Store, 2021. *arduino-uno-rev3*. [Online]
Available at: <https://store.arduino.cc/products/arduino-uno-rev3>
[Accessed 21 05 2023].
- Eda Gungor, H. M., 2020. "Light-emitting diodes (LEDs): Materials, devices, and applications". *Journal of Applied Physics*, 127(17).
- H. Z. Khatatneh, M. L. L., 2016. *Design and development of a new DC power jack interface*. s.l., 2016 IEEE International Conference on Consumer Electronics (ICCE).
- Jost, D., 2019. *what-ir-sensor*. [Online]
Available at: <https://www.fierceelectronics.com/sensors/what-ir-sensor>
[Accessed 21 05 2023].
- Macgregor, C., 2016. Fatal Attraction : how street lights prevent moths from pollinating. *The Conversation*.
- Matthews, K., 2019. 5 Ways Automation is Saving Energy Across Industries. *blue&green tomorrow*, 26 06.
- Road Safety Toolkit, 2022. *street-lighting/*. [Online]
Available at: <https://toolkit.irap.org/safer-road-treatments/street-lighting/>
[Accessed 21 05 2023].
- Robu.in , 2020. *ir-sensor-working/*. [Online]
Available at: <https://robu.in/ir-sensor-working/>
[Accessed 21 05 2023].
- Uttley, J., Monteiro, A. L. & Fotios, S., 2018. Fatal attraction: how street ligths prevent moths from pollinating. *The Conversation*.